Machine Learning Roadmap

teach me basic of machine learning ,i want to learn in 2 months complete machine learning for data scientist role and give me now 2 months roadmap with theory and coding demonstration and every day 30 mcq and each topic related case studies and potential interview question asked by top companies and answer them and first create a roadmap for 2 months in tabular formatt like table
Show moreShow less

Absolutely. Since your goal is specifically **Data Scientist**, I'd structure the 2 months around **ML fundamentals** → **classical ML** → **advanced ML** → **real-world case studies** → **model interpretation** → **production concepts** → **interview preparation**.

The plan assumes roughly **2–3 hours/day**. Every study day will follow this structure:

- **Theory:** concepts and intuition
- **Coding demonstration:** Python + NumPy/Pandas + scikit-learn
- **30 MCQs:** concept + interview-oriented questions
- **Case study:** business/data-science application
- **Interview questions:** including company-style questions
- **Practice:** implementation/exercises
- **Mini project:** where applicable

🚀 2-Month Machine Learning Roadmap for Data Scientist

Day	Topic	Theory You'll Learn	Coding / Practical	Case Study
1	Machine Learning Fundamentals	What is ML, AI vs ML vs DL, types of ML, ML workflow	First ML program with Python	Customer churn
2	ML Problem Formulation	Features, target, observations, training objective	Define ML problems	Banking default prediction
3	Dataset & Data Understanding	Numerical/categorical data, distributions, missing values	Pandas + EDA	Customer dataset
4	Train/Test Split	Training, validation, testing, data leakage	`train_test_split()`	Credit risk
5	Feature Engineering Basics	Scaling, encoding, transformations	StandardScaler, OneHotEncoder	Customer segmentation
6	Linear Algebra for ML	Vectors, matrices, dot product	NumPy implementation	Linear regression intuition
7	Statistics for ML	Mean, variance, covariance, correlation	Statistical analysis	Financial dataset
8	Probability for ML	Probability, conditional probability, Bayes	Python probability examples	Fraud detection

Day	Topic	Theory You'll Learn	Coding / Practical	Case Study
		theorem		
9	Linear Regression	Regression intuition, equation, coefficients	**`LinearRegression`**	House price prediction
10	Linear Regression Mathematics	Cost function, MSE, gradient descent	Implement regression from scratch	Sales prediction
11	Regression Evaluation	MAE, MSE, RMSE, R^2, adjusted R^2	Evaluate models	Revenue prediction
12	Multiple Linear Regression	Multiple features, multicollinearity	sklearn implementation	Salary prediction
13	Polynomial Regression	Non-linear relationships	PolynomialFeatures	Demand forecasting
14	Regularization	Ridge, Lasso, Elastic Net	sklearn implementation	High-dimensional data
15	Logistic Regression	Classification, sigmoid, probabilities	**`LogisticRegression`**	Customer churn
16	Classification Metrics	Accuracy, precision, recall, F1	Confusion matrix	Fraud detection
17	ROC-AUC & PR-AUC	Thresholds, ROC curve, AUC	**`roc_auc_score`**	Credit default
18	K-Nearest Neighbors	Distance, KNN algorithm	**`KNeighborsClassifier`**	Customer classification
19	Naive Bayes	Bayes theorem, independence assumption	Gaussian/Multinomial NB	Spam detection
20	Decision Trees	Splitting, entropy, Gini, information gain	**`DecisionTreeClassifier`**	Loan approval
21	Decision Tree Regression	Regression trees, leaf predictions	DecisionTreeRegressor	Price prediction
22	Random Forest	Bagging, bootstrap, feature randomness	**`RandomForestClassifier`**	Credit risk
23	Ensemble Learning	Bagging vs boosting vs stacking	Ensemble models	Fraud detection
24	Gradient Boosting	Sequential learning, weak learners	GradientBoostingClassifier	Customer churn
25	XGBoost	Boosting, regularization, hyperparameters	XGBoost	Credit default
26	LightGBM & CatBoost	Efficient boosting, categorical features	LightGBM/CatBoost	Financial prediction
27	Model Comparison	Linear vs tree vs ensemble models	Compare multiple models	End-to-end classification
28	**Week 4 Project**	ML workflow revision	Complete ML pipeline	**Credit Risk Prediction**
29	Unsupervised Learning	Supervised vs unsupervised learning	Clustering introduction	Customer segmentation
30	K-Means Clustering	Centroids, distance, iterations	**`KMeans`**	Customer segmentation

Day	Topic	Theory You'll Learn	Coding / Practical	Case Study
31	Choosing K	Elbow method, silhouette score	Cluster evaluation	Marketing segmentation
32	Hierarchical Clustering	Agglomerative clustering, dendrograms	**`AgglomerativeClustering`**	Customer groups
33	DBSCAN	Density-based clustering, noise	DBSCAN implementation	Fraud/outlier detection
34	PCA	Dimensionality reduction, variance	**`PCA`**	High-dimensional customer data
35	PCA Mathematics	Eigenvectors, eigenvalues, projections	PCA from intuition	Financial features
36	Anomaly Detection	Outliers, Isolation Forest	**`IsolationForest`**	Transaction fraud
37	Feature Selection	Filter, wrapper, embedded methods	SelectKBest, RFE	Credit scoring
38	Cross-Validation	K-fold, stratified K-fold, validation	**`cross_val_score`**	Model selection
39	Hyperparameter Tuning	Parameters vs hyperparameters	GridSearchCV	Random Forest tuning
40	Randomized Search & Optuna	Efficient optimization	RandomizedSearchCV	XGBoost optimization
41	Bias-Variance Tradeoff	Underfitting, overfitting	Learning curves	Model diagnosis
42	Regularization Deep Dive	L1/L2, model complexity	Compare regularized models	High-dimensional ML
43	Feature Engineering	Binning, ratios, log transformations	Feature pipeline	Financial data
44	Handling Imbalanced Data	Oversampling, undersampling, SMOTE	**`imbalanced-learn`**	Fraud detection
45	Data Leakage	Leakage sources and prevention	Leakage demonstration	Credit risk
46	Pipelines	Transformers + estimator	sklearn Pipeline	Production-style ML
47	ColumnTransformer	Numerical + categorical pipelines	ColumnTransformer	Banking dataset
48	Explainable ML	Feature importance, coefficients	Feature importance	Loan approval
49	SHAP	SHAP values, local/global explanation	SHAP implementation	Credit risk
50	LIME	Local explanations	LIME implementation	Customer churn
51	Model Calibration	Probability calibration, Brier score	CalibrationCurve	Default probability
52	Threshold Optimization	Business-driven classification thresholds	Optimize precision/recall	Fraud detection
53	Time Series ML Basics	Time-dependent data, temporal split	Lag features	Stock/demand data
54	ML for Time Series	Lag/rolling features	Random Forest/XGBoost	Sales forecasting

Day	Topic	Theory You'll Learn	Coding / Practical	Case Study
55	Survival Analysis Basics	Time-to-event concepts	Survival analysis intro	Customer churn
56	**Week 8 Project**	Full ML lifecycle	Complete project	**Fraud Detection System**
57	ML System Design	ML architecture, training/inference	Design ML system	Recommendation system
58	Model Deployment Basics	Serialization, APIs, inference	Joblib + FastAPI/Flask	Churn API
59	ML Monitoring	Data drift, concept drift, model performance	Monitoring concepts	Production credit model
60	**Final Data Scientist Project + Interview Day**	Full revision	End-to-end ML project	**Enterprise ML Case Study**

📚 What You'll Know After 60 Days

The roadmap is deliberately broader than simply learning `scikit-learn`.

1. Mathematical foundations

You'll learn the mathematics needed to **understand ML rather than memorize APIs**:

- Linear algebra
- Probability
- Statistics
- Optimization
- Cost/loss functions
- Gradient descent
- Bias-variance tradeoff

2. Supervised Learning

You'll become comfortable with:

Regression

- Linear Regression
- Multiple Regression
- Polynomial Regression
- Ridge
- Lasso
- Elastic Net

Classification

- Logistic Regression
- KNN
- Naive Bayes
- Decision Trees
- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost

---

3. Unsupervised Learning

You'll cover:

- K-Means
- Hierarchical Clustering
- DBSCAN
- PCA
- Anomaly Detection
- Isolation Forest

These are especially useful for **customer segmentation, fraud detection and exploratory analysis**.

---

4. Model Evaluation

This is one of the most important areas for Data Scientist interviews.

You'll learn when to use:

Regression

$$MAE$$

$$MSE$$

$$RMSE$$

$$R^2$$

Classification

$$Accuracy$$

$$Precision$$

$$Recall$$

$$F1$$

$$ROC - AUC$$

$$PR - AUC$$

and importantly:

"Which metric should I choose for this business problem?"

For example:

> In fraud detection, accuracy can be misleading because fraudulent transactions are usually a tiny percentage of all transactions.

🧠 5. Advanced Data Scientist Skills

We'll go beyond basic ML tutorials.

You'll learn:

- Cross-validation
- Hyperparameter optimization
- Grid Search
- Random Search
- Optuna
- Feature selection
- Feature engineering
- Class imbalance
- SMOTE
- Data leakage
- Pipelines
- ColumnTransformer
- Model calibration
- Threshold optimization
- SHAP
- LIME
- Model interpretability

🏦 6. Case Studies

Because you're targeting a **Data Scientist role**, the case studies will be business-oriented rather than just "predict iris flowers."

We'll repeatedly use realistic domains:

Domain	ML Problem
🏦 Banking	Credit default prediction
💳 Finance	Fraud detection
📱 Telecom	Customer churn
🛒 E-commerce	Customer segmentation
📈 Finance	Stock/financial prediction
🏠 Real Estate	Price prediction
🏭 Manufacturing	Predictive maintenance
📦 Retail	Demand forecasting
🎬 Entertainment	Recommendation systems
💰 Marketing	Customer response prediction

💻 Coding Stack

The main stack will be:

```
Python
│
├── NumPy
├── Pandas
├── Matplotlib
├── Seaborn
│
└── Scikit-learn
      │
      ├── Preprocessing
      ├── Regression
      ├── Classification
      ├── Clustering
      ├── Dimensionality Reduction
      ├── Model Selection
      ├── Metrics
      └── Pipelines
```

Then:

```
Advanced ML
│
├── XGBoost
├── LightGBM
├── CatBoost
├── SHAP
├── LIME
└── Imbalanced-learn
```

And eventually:

```
Production
│
├── Joblib
```

```
├── FastAPI
├── Docker concepts
├── Model monitoring
└── Data/Model drift
```

🎯 Daily Lesson Format

Every day, I'll teach you using a consistent structure.

Part 1 — Concept

For example:

> What is Logistic Regression?

We'll cover:

intuition → mathematics → assumptions → algorithm → implementation → interpretation → business application

---

Part 2 — Coding Demonstration

We'll start simple:

```python
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()

model.fit(X_train, y_train)

predictions = model.predict(X_test)
```

Then progressively move toward production-quality code:

```python
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer

pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("model", LogisticRegression())
])

pipeline.fit(X_train, y_train)
```

🧪 Part 3 — 30 MCQs Every Day

Each lesson will contain **30 questions**.

Example:

Q1. Which metric is generally more appropriate than accuracy for highly imbalanced fraud data?

A. Accuracy
B. Recall

C. R^2
D. MSE

Answer: B — Recall

Then I'll explain **why the answer is correct and why the other options are wrong**.

You'll therefore get:

60 × 30 = 1,800 MCQs

over the complete roadmap.

🏢 Part 4 — Interview Questions

Every lesson will also include interview questions.

We'll divide them into:

Beginner

> What is supervised learning?

Intermediate

> What is the difference between bagging and boosting?

Advanced

> Why can Random Forest reduce variance compared with a single decision tree?

Data Scientist

> Your fraud model has 99.5% accuracy. Is this a good model?

Case-study interview

> A bank wants to predict loan defaults. How would you build the model from raw data through deployment?

Senior-level thinking

> Your model performs extremely well offline but poorly after deployment. How would you investigate the issue?

🏆 Interview Preparation

We'll particularly train for the type of questions commonly encountered in Data Scientist interviews at large technology, consulting, e-commerce and financial-services companies.

The focus won't just be:

> "What is Random Forest?"

It will be:

> **Why would you choose Random Forest over Logistic Regression for this problem?**

and:

> **What would you do if the model has high training accuracy but low validation accuracy?**

and:

> **How would you explain the model to a non-technical business stakeholder?**

That distinction is extremely important for a Data Scientist role.

🔥 The 4 Major Projects

By the end of the 60 days, you should have at least these four portfolio-quality projects:

Project 1 — Credit Risk Prediction

Business problem:

Predict whether a customer will default on a loan.

You'll implement:

```
EDA
 ↓
Data Cleaning
 ↓
Feature Engineering
 ↓
Train/Test Split
 ↓
Preprocessing
 ↓
Baseline Model
 ↓
Tree Models
 ↓
XGBoost
 ↓
Hyperparameter Tuning
 ↓
SHAP
 ↓
Threshold Optimization
 ↓
Business Recommendation
```

Project 2 — Customer Churn Prediction

Predict customers likely to leave a telecom/subscription service.

You'll learn how to translate:

ML output → business action

For example:

> "Which customers should the company target with retention offers?"

Project 3 — Fraud Detection

You'll handle:

- Extreme class imbalance
- Precision/recall tradeoffs
- PR-AUC
- SMOTE
- anomaly detection
- threshold optimization
- false positives vs false negatives

This is particularly valuable for **finance/risk Data Scientist roles**.

Project 4 — End-to-End Enterprise ML System

The final project combines:

```
Raw Data
    ↓
EDA
    ↓
Feature Engineering
    ↓
ML Pipeline
    ↓
Model Training
    ↓
Hyperparameter Tuning
    ↓
Evaluation
    ↓
SHAP
    ↓
Model Serialization
    ↓
API
    ↓
Monitoring
```

This becomes a strong **Data Scientist portfolio project**.

📈 Your Progression

Think of the 60 days as:

```
WEEK 1
ML + Math Foundations
         ↓
```

```
WEEK 2
Regression
        ↓
WEEK 3
Classification
        ↓
WEEK 4
Ensemble Learning
        ↓
WEEK 5
Unsupervised ML
        ↓
WEEK 6
Model Optimization
        ↓
WEEK 7
Advanced ML + Explainability
        ↓
WEEK 8
Production + Projects + Interviews
```

By the end, the target is not simply:

> **"I know machine learning algorithms."**

The target is:

> **"I can take a business problem, convert it into an ML problem, prepare the data, engineer features, select and train models, evaluate them correctly, explain the model, optimize it, and communicate the business impact."**

That is the skill set we'll build toward for a **Data Scientist role**.